

SUPPLEMENTARY MATERIAL: LAU-GRES DATASET AND LEAD FRAMEWORK

Overview. This supplement expands the dataset construction, experimental protocol, training objectives, and diagnostic analyses presented in the main paper. Unless otherwise stated, the additional ablations are conducted on LAU-GRES, and the reported values are averaged over the validation and test splits.

1. DATASET CONSTRUCTION DETAILS

1.1. Source Imagery and Patch Extraction

LAU-GRES is constructed from CODrone [1], a low-altitude UAV oriented-object detection benchmark. CODrone images have a resolution of 3840×2160 and cover urban roads, parking areas, residential regions, ports, and other scenes with substantial viewpoint and illumination variation. Its rotated bounding boxes provide tighter object prompts than horizontal boxes, particularly for elongated targets and instances observed from oblique views.

Each source image is uniformly divided into 1080×1080 patches. Cropping native-resolution pixels instead of globally downsampling the complete frame preserves the appearance of pedestrians, bicycles, and other small targets. The resulting patches are manually filtered to remove samples dominated by empty background, severe blur, duplicated content, or unusable occlusion. The retained subset emphasizes oblique downward views of approximately 30° – 60° and includes challenging nighttime scenes.

Eight frequent object categories are retained: pedestrian, bicycle, tricycle, car, motorcycle, truck, bus, and boat. The taxonomy covers common urban mobility objects while preserving difficult distinctions such as bicycle versus motorcycle and car versus truck.

The construction pipeline consists of six stages. We first select representative CODrone frames and their rotated-box annotations. Each frame is then cropped into 1080×1080 patches, followed by manual filtering of low-quality and redundant content. SAM-2 generates instance masks from the rotated-box prompts. A multimodal large language model subsequently generates generalized referring expressions, and human annotators finally verify and correct every image-expression-mask triplet.

1.2. Automatic Mask Generation

Rotated boxes supplied by CODrone are used as prompts for SAM-2 [2] to obtain instance-level masks. Crowded UAV scenes frequently contain adjacent objects with similar color, scale, and orientation. Therefore, the automatic masks are inspected for boundary adhesion, missing thin structures, under-segmentation, and leakage into neighboring instances. Samples whose masks cannot be corrected reliably are discarded.

The prompt-based procedure substantially reduces annotation cost while retaining accurate object boundaries. Rotated prompts are particularly useful for vehicles under oblique viewpoints because they reduce the amount of irrelevant background within the prompt region.

1.3. Generalized Referring Expression Generation

For expression generation, the multimodal language model receives two complementary views. The first is the complete image with the target region indicated, preserving spatial and relational context. The second is an upsampled crop of the same target, revealing fine-grained attributes that are difficult to identify at the original UAV scale. The model is instructed to generate concise English expressions without mentioning boxes, identifiers, annotations, or the generation process.

Three referring scenarios are considered. A *single-target* expression identifies one instance. A *multi-target* expression jointly refers to several instances satisfying the same semantic condition. A *no-target* expression describes a plausible category or attribute combination that is absent from the image. LAU-GRES consequently evaluates pixel localization, multi-instance completeness, and rejection of unsupported queries.

1.4. Prompt Design for Language Annotation

We use one task-aware instruction for all samples; the placeholders *target list* and *negative category* are filled from the verified instance annotations. The consolidated prompt is given below.

Prompt: Provide two identical low-altitude drone images: the first shows the area used to identify targets from {target_list}, and the second set of images is an upsampled, clear observation image. Please generate 3 short and accurate English descriptions for the targets. If {target_list} is empty, describe {negative_category}. Do not mention boxes, labels, or annotations. Specific requirements are

as follows: Determine the task from the list. For one instance, refer only to that target and distinguish it from same-category distractors. For multiple same-category instances, every expression must cover all and only the listed instances, use plural or collective wording, and state the exact quantity whenever a number is used. For cross-category instances, cover every listed target and connect the categories with “and,” “together with,” or equivalent wording. Never omit, add, merge, or split listed targets incorrectly. Use only clearly visible category, quantity, color, size, appearance, state, orientation, image position, or relations to roads, buildings, vehicles, pedestrians, vegetation, and nearby objects. Do not invent attributes or infer identity, age, gender, occupation, or intention. Make the three expressions structurally distinct, preferably emphasizing position, appearance or state, and spatial context. If the target list is empty, inspect the complete image. If the negative category occurs anywhere, mark the sample invalid and produce nothing; otherwise generate three normal positive-style expressions for that category without revealing its absence. Do not use “no,” “not,” “absent,” “missing,” “nonexistent,” “there is no,” “cannot be found,” or “does not exist,” and do not confuse similar objects with the negative category. Never mention boxes, labels, identifiers, annotations, coordinates, pixels, candidate-box colors, or the generation process; avoid vague wording, synonym-only variants, explanations, and additional remarks. Output only the three expressions.

1.5. Human-in-the-Loop Refinement and Data Split

Annotators jointly inspect every image, expression, and segmentation mask. They correct inaccurate boundaries, remove descriptions with ambiguous referents, and revise wording that depends on unavailable context. Particular attention is paid to relative directions under camera rotation and to category names for visually similar small objects. This procedure verifies cross-modal consistency rather than evaluating the image, text, and mask independently.

The resulting dataset contains 13,369 aligned image–expression–mask triplets. It is randomly divided into training, validation, and test subsets using a 7:1:2 ratio. The split is fixed across all experiments. The validation subset is used for model selection, while the held-out test subset is used for final comparisons.

1.6. Dataset Organization

LAU-GRES separates image assets, instance-level annotations, and generalized referring annotations to avoid duplicating pixel masks across expressions. Table 1 summarizes the released directory structure and the role of each component.

The two JSON files form a relational annotation layer. Each referring record first identifies an image in `instances.json` and then associates its expression with a set of instance identifiers. A singleton set defines a single-target query, multiple identifiers define a multi-target query, and an empty set defines a no-target query. The final ground-truth mask is recovered as the union of all referenced instance masks. This organization preserves instance-level supervision

Path	Function
<code>README.md</code>	Documents the category vocabulary, directory layout, split protocol, and data-loading procedure.
<code>images/</code>	Stores the low-altitude image patches used by all referring samples. Numeric filenames provide stable keys for annotation lookup.
<code>LAU-GRES/instances.json</code>	Contains COCO-style image metadata, categories, instance identifiers, bounding boxes, areas, and segmentation masks.
<code>LAU-GRES/LAU-GRES_grefs_full.json</code>	Stores generalized referring records, including the expression, image association, split, and the zero, one, or multiple target-instance identifiers.

Table 1. Directory organization of the LAU-GRES release. Paths are relative to the dataset root.

while allowing multiple linguistic queries to reuse the same image and mask annotations. It also enables deterministic reconstruction of image–expression–mask triplets without storing redundant raster masks.

1.7. Annotation Integrity and Consistency Checks

We apply three levels of validation before fixing the released annotations. First, referential-integrity checks verify that every referring record resolves to an existing image and that every target identifier resolves to an instance annotation in the same image. Identifiers must be unique within an image, a single-target record must contain exactly one identifier, and a multi-target record must contain at least two. A no-target record is required to have an empty target list. These constraints prevent silent mismatches when training samples are assembled from the two JSON files.

Second, mask-level checks validate image dimensions, polygon or run-length encodings, nonzero areas for positive instances, and agreement between stored areas and decoded masks. The generalized ground truth is independently reconstructed by taking the set union of all referenced instance masks. Its spatial extent is checked against the image boundary, and its target count is compared with the number of referenced instances. Set union is used rather than area addition so that overlapping masks cannot be counted twice.

Third, language-level checks examine the semantic consistency of each expression. Quantity words and plural forms are compared with the target-list cardinality; category names are checked against the selected instances; and claims about color, relative position, and nearby landmarks are manually verified in the image. Expressions containing annotation artifacts such as box colors, identifiers, or pixel coordinates are

RefCOCO+.

RefCOCO+ follows the same image source and collection paradigm but prohibits absolute spatial words such as *left* and *right*. It contains 141,564 expressions for 49,856 objects across 19,992 images. Localization must therefore rely more strongly on category, appearance, and relational cues, making RefCOCO+ a useful benchmark for testing whether latent expressions capture attributes beyond explicit position.

RefCOCOg.

RefCOCOg [5] was collected in a non-interactive setting, resulting in longer and more descriptive expressions. It contains 95,010 expressions for 49,822 objects across 25,799 images. We report the commonly used UMD validation/test results and the Google validation split where applicable. Its longer language provides a complementary test of latent semantic refinement.

LAU-GRES.

Our benchmark contains 13,369 image-expression-mask triplets under no-target, single-target, and multi-target scenarios. All supplementary ablations are conducted on LAU-GRES. Table 2 summarizes the datasets used in the experiments.

2.2. Evaluation Metrics

Mean Intersection over Union (mIoU) is the primary metric on RefCOCO, RefCOCO+, and RefCOCOg. On LAU-GRES, we report P@0.5, P@0.7, generalized IoU (gIoU), cumulative IoU (cIoU), mean Robust Recall (mRR), robust IoU (rIoU), and no-target accuracy (N-acc), following the generalized and robust referring segmentation protocols [6, 7].

P@X denotes the percentage of target-present samples whose IoU exceeds X. cIoU accumulates intersection and union over the complete evaluation set, whereas gIoU weights every sample equally. Let $\widehat{M}_i$ and M_i denote the predicted and ground-truth masks of sample i , respectively. The per-sample generalized score is

$$s_i = \begin{cases} \frac{|\widehat{M}_i \cap M_i|}{|\widehat{M}_i \cup M_i|}, & |M_i| > 0, \\ 1, & |M_i| = 0 \text{ and } |\widehat{M}_i| = 0, \\ 0, & |M_i| = 0 \text{ and } |\widehat{M}_i| > 0. \end{cases} \quad (1)$$

For an evaluation set containing N expressions, gIoU is defined as

$$\text{gIoU} = \frac{1}{N} \sum_{i=1}^N s_i. \quad (2)$$

Thus, a correctly rejected no-target expression contributes one, whereas any false foreground prediction for a no-target expression contributes zero, as defined in GRES [6].

For robust evaluation [7], let $\mathcal{R}$ be the set of image references. Reference R_i contains p_i positive expressions with

predictions $\widehat{m}_j^i$ and ground-truth masks m_j^i , together with n_i negative expressions whose predicted masks are $\widetilde{m}_k^i$. Define

$$A_i = \sum_{j=1}^{p_i} |\widehat{m}_j^i \cap m_j^i|, \quad (3)$$

$$B_i = \sum_{j=1}^{p_i} |\widehat{m}_j^i \cup m_j^i| + \sum_{k=1}^{n_i} |\widetilde{m}_k^i|. \quad (4)$$

The robust IoU jointly rewards positive-expression overlap and penalizes false foreground produced by negative expressions:

$$\text{rIoU} = \frac{1}{|\mathcal{R}|} \sum_{i=1}^{|\mathcal{R}|} \frac{A_i}{B_i}. \quad (5)$$

For the same reference, Robust Recall measures the fraction of negative expressions that produce an exactly empty mask:

$$\text{RR}_i = \frac{1}{n_i} \sum_{k=1}^{n_i} \mathbb{I}(|\widetilde{m}_k^i| = 0), \quad (6)$$

$$\text{mRR} = \frac{1}{|\mathcal{R}|} \sum_{i=1}^{|\mathcal{R}|} \text{RR}_i, \quad (7)$$

where $\mathbb{I}(\cdot)$ is the indicator function. N-acc separately reports the proportion of no-target samples for which the model predicts an empty mask. Unlike mRR, which is first averaged within each reference, N-acc treats all no-target expressions as independent samples.

2.3. Metric Interpretation and Edge Cases

The metrics emphasize complementary aspects of generalized referring segmentation. cIoU pools foreground pixels over the dataset and is therefore more influenced by large objects, whereas gIoU assigns equal weight to every expression and exposes performance on small referents. For no-target samples, gIoU is intentionally discrete: an exactly empty prediction is correct, while any false foreground is incorrect. This convention prevents a tiny spurious mask from receiving an artificially favorable overlap score.

rIoU evaluates positive and negative expressions jointly within an image reference. False foreground from negative expressions enlarges its denominator even when positive masks are localized correctly. mRR is stricter in a different way: it records whether each negative query is rejected completely, independent of the number of false pixels. Consequently, an always-empty model could obtain high negative-query recall but would fail gIoU, cIoU, and rIoU on target-present samples; conversely, an over-segmenting model could retain moderate positive overlap while obtaining poor mRR and N-acc. Reporting the metrics together therefore rules out these degenerate solutions.

Dataset	Images	Objects	Expressions	Key characteristic
RefCOCO	19,994	50,000	142,210	Short expressions; spatial cues allowed
RefCOCO+	19,992	49,856	141,564	Absolute spatial terms prohibited
RefCOCog	25,799	49,822	95,010	Longer non-interactive descriptions
LAU-GRES	–	–	13,369	Low-altitude generalized segmentation

Table 2. Summary of the datasets used in the experiments. LAU-GRES is reported in aligned image–expression–mask triplets.

All probability maps are binarized with the same fixed threshold used in the main experiments. Empty-mask decisions are made after resizing the prediction to the evaluation resolution, and no per-dataset or per-category threshold is tuned on the test set. This protocol ensures that differences in negative-query robustness reflect model behavior rather than threshold selection.

The same image resizing, text preprocessing, and output resolution are used for all comparison methods whenever their public implementations permit. No LAU-GRES test annotation is used for optimization or hyperparameter selection.

2.4. Loss Functions

Let $P \in [0, 1]^{H \times W}$ denote the predicted foreground probability map and $Y \in \{0, 1\}^{H \times W}$ the ground-truth mask. The pixel-wise binary cross-entropy loss is

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{HW} \sum_{j=1}^{HW} [Y_j \log P_j + (1 - Y_j) \log(1 - P_j)]. \quad (8)$$

To alleviate foreground–background imbalance, we also use the soft Dice loss

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_j P_j Y_j + \epsilon}{\sum_j P_j + \sum_j Y_j + \epsilon}, \quad (9)$$

where ϵ is a numerical-stability constant. The complete mask objective is

$$\mathcal{L}_{\text{mask}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}. \quad (10)$$

The Counting-Aware Head predicts a category-specific count vector $\mathbf{C}_{\text{pred}}$ and is supervised by $\mathbf{C}_{\text{gt}}$ using

$$\mathcal{L}_{\text{count}} = \frac{1}{K} \sum_{k=1}^K \text{SmoothL1}(C_{\text{pred},k} - C_{\text{gt},k}). \quad (11)$$

Here, $\text{SmoothL1}(x) = 0.5x^2$ for $|x| < 1$ and $|x| - 0.5$ otherwise. It is less sensitive to occasional large counting errors than an ℓ_2 objective while retaining a smooth gradient near the optimum.

A lightweight classifier receives a stop-gradient copy of the predicted count vector and estimates target existence. For binary label y and predicted probability q , the existence objective is

$$\mathcal{L}_{\text{exist}} = -[y \log q + (1 - y) \log(1 - q)]. \quad (12)$$

Component	Configuration
Visual encoder	Pretrained Swin-B
Text encoder	Pretrained BERT
Hierarchical decoder	Three cascaded SIM stages
Feature dimension	256
Multi-head attention	8 heads; 32 dimensions per head
Feed-forward network	Hidden dimension of 1024
Decoder dropout	0.1
Visual token grid	30×30
Latent expressions	$N = 2$; token lengths $\{4, 6\}$
Mask probabilities	$\{0.15, 0.20\}$
Orthogonal anchors	$N_c = 100$
Image resolution	1080^2 crops; 480^2 model input
Optimizer	AdamW; end-to-end training
Batch size / hardware	8 / one NVIDIA RTX 5090
Loss weights	$\lambda_1 = 1, \lambda_2 = 0.5, \lambda_3 = 1$
Data split	7:1:2

Table 3. Default implementation settings used for LEAD.

Stopping its gradient prevents the coarse binary objective from collapsing the richer category-wise count representation. The complete objective is

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{mask}} + \lambda_2 \mathcal{L}_{\text{count}} + \lambda_3 \mathcal{L}_{\text{exist}}, \quad (13)$$

where $\lambda_1 = 1$, $\lambda_2 = 0.5$, and $\lambda_3 = 1$.

2.5. Implementation Details

LEAD uses ImageNet-pretrained Swin-B [8] and an officially pretrained BERT model [9] as the visual and textual encoders, respectively. Three cascaded Semantic Interaction Modules form the hierarchical decoder. Multi-scale features are projected to 256 dimensions. Attention uses eight heads with 32 dimensions per head, and the feed-forward layer has a hidden dimension of 1024. The pooled visual-token grid is 30×30 .

Two latent expressions are used, with dropout-mask probabilities $\{0.15, 0.20\}$, token lengths $\{4, 6\}$, and 100 orthogonal anchor tokens. Images are resized to 480×480 before entering the visual encoder. The network is trained end-to-end with AdamW [10], batch size 8, on one NVIDIA GeForce RTX 5090. Unless one setting is explicitly changed in an ablation, every other configuration remains fixed.

3. ADDITIONAL EXPERIMENTS

3.1. Comparison Scope and Baseline Families

The comparison methods cover several representative design families. LAVT introduces language-aware visual features through early transformer fusion [11], while DMMI uses multiple forms of cross-modal interaction and iterative mask refinement [12]. ReLA models regional linguistic alignment for generalized referring segmentation [13], and InstAlign extends this direction with instance-aware matching [14]. CoHD organizes the decoder as a counting-aware hierarchy [15]. These methods provide strong references for determining whether LEAD’s improvements arise from latent semantic refinement, hierarchical fusion, or explicit count supervision.

Earlier RIS systems provide complementary architectural references. VLT uses language-conditioned query generation to obtain multiple visual interpretations [16], while CRIS transfers CLIP knowledge through vision–language decoding and text-to-pixel contrastive alignment [17]. PolyFormer instead formulates segmentation as autoregressive polygon generation [18]. Together, these methods broaden the comparison context from conventional fusion decoders to pre-trained vision–language alignment and geometric sequence prediction.

We additionally consider recent generalized systems built on larger visual or multimodal models. GSVA formulates generalized segmentation with multimodal large language models [19], whereas DeRIS separates perception and cognition to improve referring-mask reasoning [20]. PropVG represents a proposal-driven grounding family with multi-granularity discrimination [21]. Latent-VG is the closest conceptual reference because it learns latent expressions for both grounding and segmentation [22]. Together, these baselines span single-encoder, region-alignment, counting-aware, proposal-driven, latent-expression, and multimodal-language model paradigms.

Whenever released configurations permit, evaluation uses the authors’ public implementation, the official dataset split, the same input resolution, and a fixed mask threshold. Comparisons with Swin-B/BERT methods are the most controlled because they share encoder capacity with LEAD. Results from systems using substantially larger pretrained backbones remain informative but are interpreted as end-to-end system comparisons rather than architecture-only ablations. No baseline is tuned using LAU-GRES test annotations. We retain each method’s standard training recipe unless a change is necessary to support no-target or multi-target samples, in which case the adaptation is applied uniformly and reported with the corresponding result.

3.2. Loss-Weight Analysis

We vary the mask and count coefficients while fixing the existence-loss weight. Count-aware supervision has also

λ_{mask}	λ_{count}	gIoU	cIoU	N-acc	C-acc
1	1	58.72	63.26	83.94	74.86
2	1	58.39	63.45	84.21	75.19
2	0.5	58.88	62.97	83.76	75.94
1	0.2	58.56	63.58	84.33	75.56
1	0.5	59.13	63.81	84.68	76.28

Table 4. Ablation with different mask/count loss weights. All values are reported in percentage.

proved effective for separating single-, multiple-, and no-target cases in CoHD [15]. As reported in Table 4, the most balanced setting is $\lambda_{\text{mask}} = 1$ and $\lambda_{\text{count}} = 0.5$. It reaches 59.13% average gIoU, 63.81% average cIoU, 84.68% average N-acc, and 76.28% count accuracy (C-acc).

Increasing the mask coefficient to 2 does not improve gIoU, while equal mask and count weights place excessive emphasis on the count branch. Conversely, a very small count coefficient weakens count-sensitive discrimination. The selected ratio provides the best balance among foreground segmentation, no-target rejection, and category-wise count prediction.

3.3. Number of SIM Stages

The hierarchy is varied from two to four stacked SIM stages. This analysis is motivated by prior hierarchical decoding [15] and latent-expression modeling [22], while isolating the number of interaction levels in our architecture. Figure 4 shows that three stages consistently achieve the strongest results: 59.13% gIoU, 63.81% cIoU, 84.68% N-acc, 92.71% mRR, and 53.88% rIoU. Two stages provide insufficient refinement, whereas four stages introduce redundant transformations that reduce positive-mask quality and negative-expression robustness.

The result supports the intended coarse-to-fine interpretation. The first SIM provides broad localization, the second resolves intermediate object structure, and the third sharpens instance boundaries. A fourth stage does not introduce a distinct semantic granularity and may amplify noise or overfit the training distribution.

The simultaneous increase in gIoU and N-acc indicates that the selected depth does not improve overlap by simply predicting more foreground. Higher mRR and rIoU further show that hierarchical refinement suppresses unsupported responses under negative expressions. We therefore use three SIM stages in all remaining experiments.

3.4. Robustness Across Target Scales

Figure 5 partitions target-present samples according to their mask-area ratios. All methods improve rapidly from the extremely small 0–1% bin to medium-sized targets, confirming that tiny objects are a major source of error. Unlike the

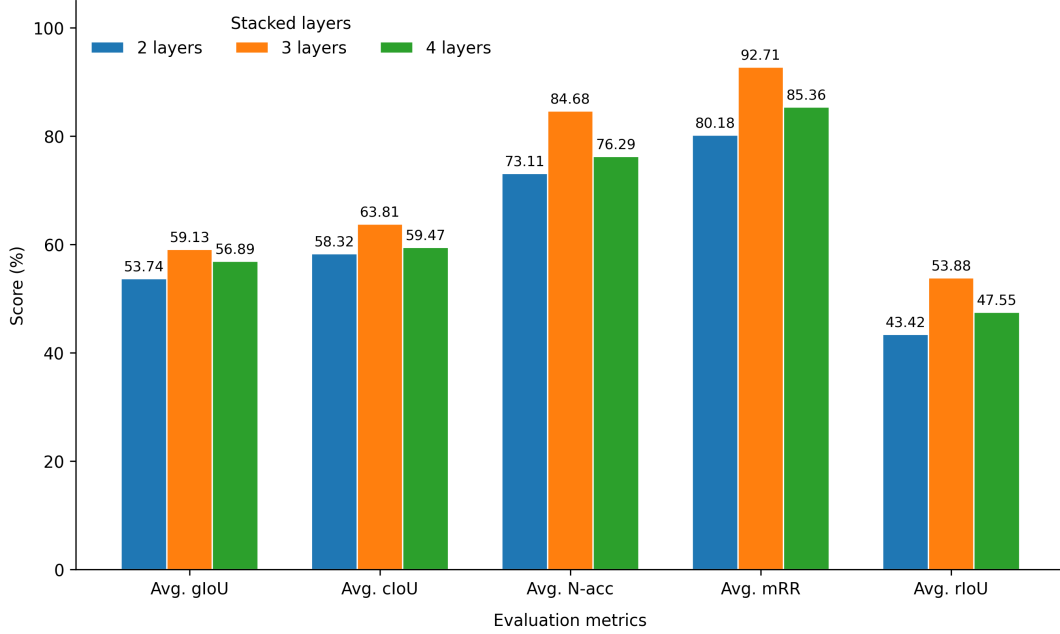


Fig. 4. Performance comparison using two, three, and four stacked SIM stages. The three-stage hierarchy achieves the best result for every reported metric.

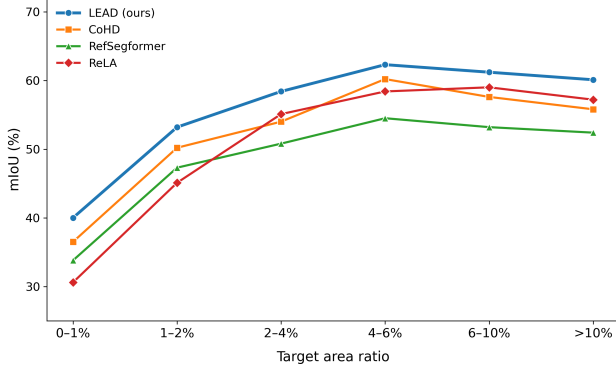


Fig. 5. Scale-wise mIoU comparison on LAU-GRES. Samples are grouped according to their target-mask area ratios.

natural-image GRES setting [6], LAU-GRES places most referents in the small-object regime. LEAD obtains the highest mIoU in every interval, including the difficult 0-1% bin, where visual evidence is sparse and nearby same-category instances are easily confused.

The advantage is not limited to one scale. LEAD continues to lead between 1% and 10%, showing that latent attribute refinement and adaptive fusion remain useful after the target becomes visually resolvable. Its curve is also more stable in the two largest bins, whereas several baselines reach a peak and then decline.

Scale-wise evaluation separates improvements on gen-

AHF	CAH	gIoU	cIoU	N-acc
×	×	46.58	55.63	63.42
✓	×	49.94	58.22	67.29
×	✓	55.25	60.47	80.06
✓	✓	59.13	63.81	84.68

Table 5. Individual and joint effects of AHF and CAH on LAU-GRES.

uinely small objects from gains dominated by a few large masks. Since approximately 79% of LAU-GRES samples occupy less than 4% of the image, performance in the first three bins directly influences the aggregate result. LEAD’s consistent advantage therefore supports its design goal rather than reflecting a favorable scale distribution.

3.5. Complementarity of AHF and CAH

Adaptive Hierarchical Fusion (AHF) and the Counting-Aware Head (CAH) address different failure modes. AHF combines complementary semantic responses and improves boundary recovery, whereas CAH strengthens discrimination among no-target, single-target, and multi-target cases. This factorized analysis complements the counting-aware hierarchy studied by CoHD [15]. Their joint use yields the strongest average performance.

AHF alone improves gIoU by 3.36 points over the baseline. CAH alone adds 8.67 points and raises N-acc by 16.64

points, demonstrating the importance of explicit counting in generalized referring scenarios. Combining the modules improves gIoU, cIoU, and N-acc by 12.55, 8.18, and 21.26 points, respectively. Hierarchical fusion therefore helps recover small and fragmented foreground, while count supervision prevents salient distractors from being treated as valid referents.

3.6. Hierarchical Semantic Response Analysis

Figure 6 shows the response maps produced by the three SIM stages. The first map provides broad target localization but frequently activates a large neighborhood around the referent. This behavior helps recover weak evidence from small objects, although it also introduces background responses. The second stage concentrates activation on object-level regions and begins to separate adjacent instances. The third stage produces the sharpest structures and most closely matches the target boundaries.

For the medium-sized tricycle and tiny orange bicycle, a diffuse location prior progressively becomes a compact object response. In multi-target examples, early responses cover a broad crowd or parking region, whereas later stages separate individual pedestrians or cars. Using only the final map may miss weak instances, while using only the first map pro-

duces excessive foreground. AHF therefore learns expression-conditioned weights that combine coarse semantic coverage with fine localization.

The progression also explains the quantitative optimum in Figure 4. Two stages stop before the finest refinement, whereas a fourth stage lacks a new semantic granularity. Three complementary maps form a compact hierarchy that can be fused without excessive redundancy.

3.7. Additional Qualitative Comparisons

Figure 7 compares LEAD with ReLA [13]. In no-target cases, ReLA frequently activates a visually plausible but semantically invalid distractor, such as another vehicle when the expression requests trucks or an object that violates the stated spatial relation. LEAD suppresses these responses because its category-level count representation and existence classifier must jointly confirm that a valid target is present.

For single-target expressions, LEAD more consistently selects the complete referent and rejects neighboring objects of similar appearance. Improvements are visible for trucks, buses, boats, tricycles, and motorcycles under oblique views. These examples expose the actual error modes reduced by LEAD and complement the aggregate rIoU and N-acc results.

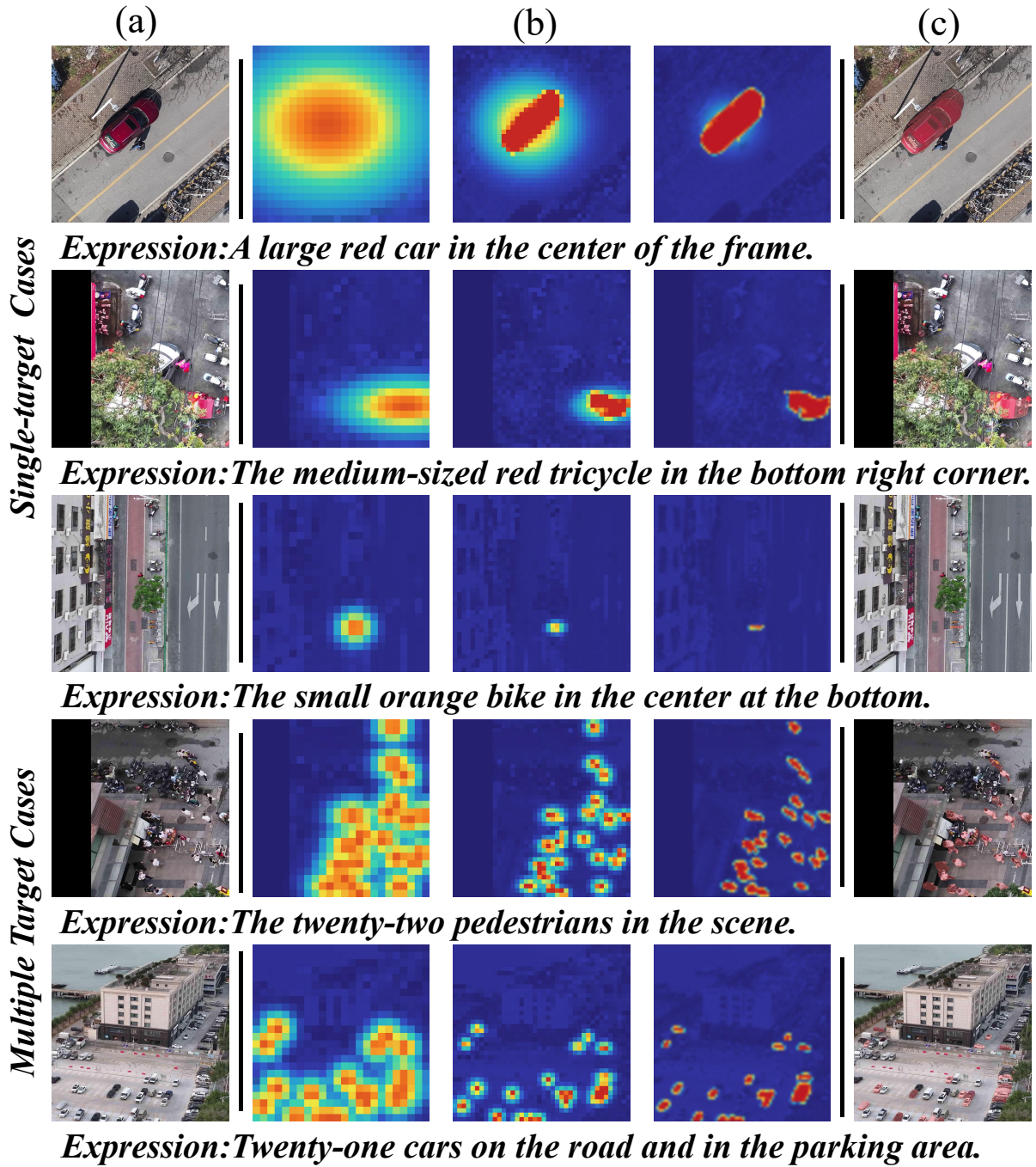


Fig. 6. Visualization of the hierarchical semantic responses produced by the three SIM stages. From left to right: (a) the input image; (b) the coarse-, intermediate-, and fine-grained response maps; and (c) the ground-truth mask. Both single-target and multi-target examples are shown.

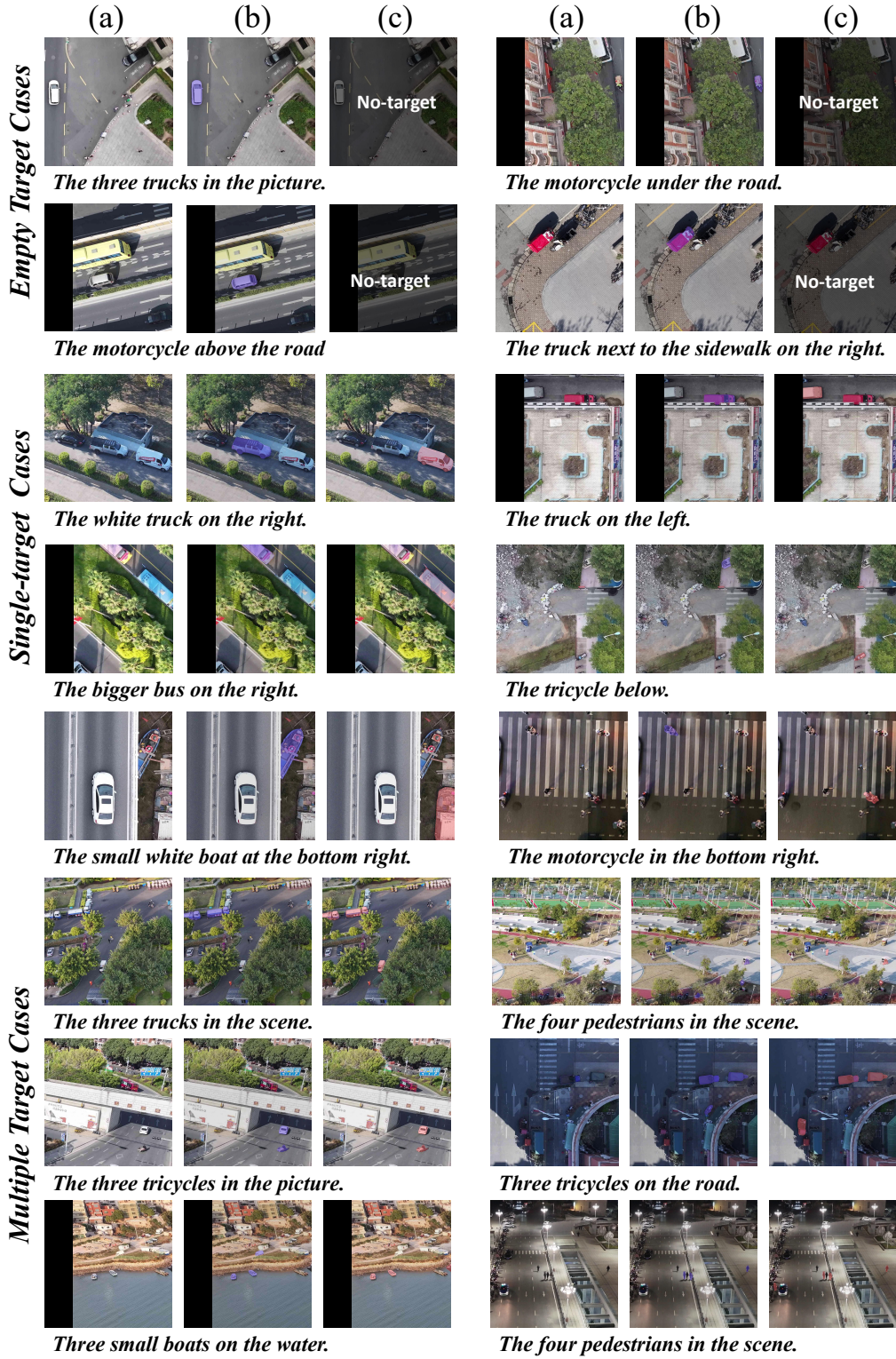


Fig. 7. Additional qualitative comparisons on LAU-GRES. Columns (a), (b), and (c) denote the input image, the ReLA prediction, and the LEAD prediction, respectively. The first two rows show no-target cases, while the remaining rows show single-target cases.

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